

From Regulatory Frameworks to System Performance: Advancing Extended Producer Responsibility for Plastic Packaging in Tanzania

Abstract



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Plastic packaging has become an essential component of Tanzania's economic development, supporting food distribution, manufacturing, healthcare, agriculture, retail trade, and consumer markets. Its affordability, durability, and convenience have contributed to improved product protection and expanded access to goods. However, increasing dependence on plastic packaging has also created a growing environmental challenge because waste management systems have not expanded at the same pace as plastic consumption.

Extended Producer Responsibility (EPR) provides a policy approach that shifts responsibility for post-consumer packaging management toward producers and importers.

The principle recognizes that companies introducing packaging into the market should contribute to the systems required to collect, recover, recycle, and manage those materials after use.

Tanzania has made progress in strengthening plastic governance through regulatory measures and environmental management frameworks. However, continued plastic leakage into public spaces, drainage systems, waterways, and coastal environments demonstrates that regulatory frameworks alone are insufficient. The central challenge is moving from policy development toward effective implementation and measurable environmental outcomes.

This article argues that Tanzania requires a performance-based EPR system focused on reducing plastic leakage, improving collection and recycling rates, encouraging sustainable packaging design, strengthening recovery infrastructure, and improving coordination among producers, government institutions, recyclers, and communities.

Introduction



Plastic packaging has become one of the most important materials supporting modern economic activities. Its low cost, durability, flexibility, and protective properties have made it essential for food and beverage distribution, pharmaceutical products, agricultural inputs, household goods, and retail markets.

For Tanzania, plastic packaging has contributed to economic development by improving product safety, reducing losses during transportation, extending shelf life, and supporting access to goods across urban and rural markets.

However, the rapid growth of plastic consumption has created environmental pressures because systems for collection, recycling, and recovery have not developed at the same pace.

The challenge is not simply the existence of plastics. The challenge is that many plastic products are designed for short-term use but remain in the environment for decades when they are not properly collected and recovered.

As urban populations grow and consumption patterns change, waste management systems face increasing pressure. Cities such as Dar es Salaam experience challenges related to waste collection, disposal capacity, and leakage of waste into drainage systems and public spaces. Poor waste management contributes to environmental pollution and can increase vulnerability to flooding when waste blocks drainage channels (World Bank, 2024).

Recent estimates reported in Tanzania indicate that the country generates approximately 14.4–20.7 million tonnes of solid waste annually, of which around 12% is plastic, implying a plastic waste stream of roughly 1.7–2.5 million tonnes per year; however, these figures should be interpreted cautiously because national waste-generation estimates vary across sources and methodologies. Other recent assessments place annual plastic waste generation at approximately 0.84–1.21 million tonnes, with less than 4% reportedly recycled (Kimaro, 2025).

More specific evidence from the packaging sector illustrates the recovery gap: in 2023, approximately 30,200 tonnes of post-consumer PET bottles were reportedly collected, but only 15,000 tonnes were recycled and exported, indicating that collection does not necessarily translate into material recovery or circularity. Earlier baseline estimates also indicate that Tanzania generated approximately 315,000 tonnes of plastic waste in 2018, with only about 40% collected and approximately 4% recycled, while an estimated 29,000 tonnes leaked into oceans, rivers and lakes (International Pollutants Elimination Network [IPEN], 2026). More recently, the Government has responded by developing the 2025–2030 Waste Management Strategy, including an ambition to collect and recycle 12.5 million tonnes of plastic waste annually. Taken together, these trends demonstrate that Tanzania's central challenge is increasingly one of system performance establishing reliable mechanisms for tracking plastic placed on the market, ensuring producer accountability, expanding collection and recovery infrastructure, integrating informal waste-sector actors, and measuring actual recycling outcomes.

Plastic pollution should therefore not be viewed only as a problem of littering or disposal behaviour. It is a systemic challenge involving production decisions, packaging design, collection systems, recycling markets, institutional capacity, and coordination among stakeholders.

This strengthens the case for moving Extended Producer Responsibility (EPR) from a predominantly framework- and guideline-oriented approach toward a measurable, performance-based system in which producer obligations are linked to verified collection, recovery, recycling and environmental outcomes.

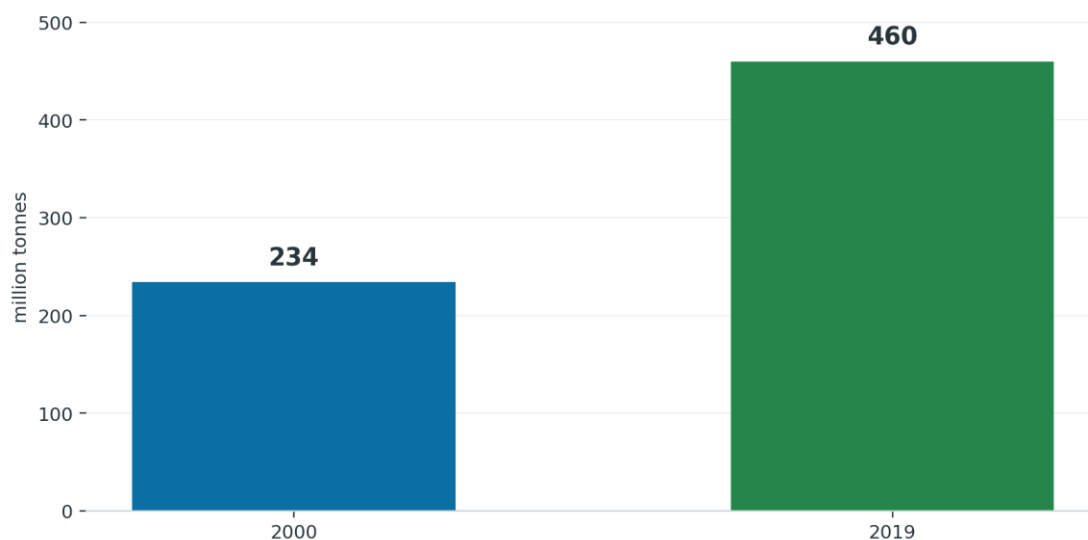
Extended Producer Responsibility provides a framework for addressing these weaknesses by ensuring that producers remain responsible for the environmental consequences of the packaging they introduce into society.

Scientific Evidence on the Environmental and Human Health Impacts of Plastic Pollution

The need for stronger plastic management systems is supported by growing scientific evidence showing that plastic pollution affects ecosystems, human health, climate systems, and economic activities.

Global plastic production has increased dramatically over recent decades. The Organisation for Economic Co-operation and Development (OECD) estimated that global plastic production increased from approximately 234 million tonnes in 2000 to 460 million tonnes in 2019.

During the same period, plastic waste generation more than doubled, demonstrating the growing gap between plastic production and the capacity of waste management systems to safely recover materials (OECD, 2022).



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One of the most significant characteristics of plastic pollution is persistence. Unlike organic materials that naturally decompose, many plastics fragment into smaller particles known as microplastics and nanoplastics. These particles can remain in ecosystems for long periods and have been detected in oceans, rivers, agricultural soils, food chains, and atmospheric environments.

Plastic pollution creates direct risks to biodiversity. Large plastic items can cause physical harm through ingestion, suffocation, and entanglement of wildlife. Marine species including fish, seabirds, turtles, and marine mammals have been documented consuming plastic materials, which can result in injury, reduced feeding ability, reproductive impacts, and mortality (Wilcox et al., 2015; Gallo et al., 2018).

The ecological impacts are particularly important for countries such as Tanzania, where coastal and marine ecosystems contribute significantly to livelihoods, fisheries, tourism, and food security. Plastic materials transported through rivers and drainage systems can eventually reach coastal environments, affecting marine habitats and species.



Scientific concerns are also increasing regarding human exposure to microplastics. Microplastic particles have been detected in drinking water, seafood, agricultural environments, and human biological samples.

Plastic pollution also contributes to climate change throughout the plastic life cycle. Fossil fuel extraction, plastic production, transportation, and disposal processes generate greenhouse gas emissions. UNEP has highlighted that addressing plastic pollution requires reducing unnecessary plastic production, improving reuse and recycling systems, and transitioning toward circular economy models (UNEP, 2023).

The scientific evidence demonstrates that plastic pollution is not only a waste management issue. It is an environmental, public health, and economic challenge that requires action across the entire plastic value chain. This strengthens the justification for EPR systems that prevent leakage, improve recovery, and encourage producers to design packaging with environmental impacts in mind.

The graphic is a vibrant, multi-layered illustration of Extended Producer Responsibility (EPR). At the center, the letters 'EPR' are rendered in large, bold, multi-colored font. Surrounding this central text are several interconnected diagrams and text blocks. A circular diagram at the top left shows the 'Circular Economy Loop' with stages: Production, Distribution, Use, and Recycling. To the left, a 'Product Lifecycle' diagram shows a flow from 'Product Development' to 'Production' to 'Distribution' to 'Use' to 'End-of-life management'. Below this, a 'Product Stewardship' diagram shows a flow from 'Product Development' to 'Production' to 'Distribution' to 'Use' to 'End-of-life management'. To the right, a 'Product Stewardship' diagram shows a flow from 'Product Development' to 'Production' to 'Distribution' to 'Use' to 'End-of-life management'. At the bottom, a 'Product Stewardship' diagram shows a flow from 'Product Development' to 'Production' to 'Distribution' to 'Use' to 'End-of-life management'. The background is a blurred image of a modern building with large windows, suggesting a corporate or industrial setting. The overall theme is sustainability and responsible production.

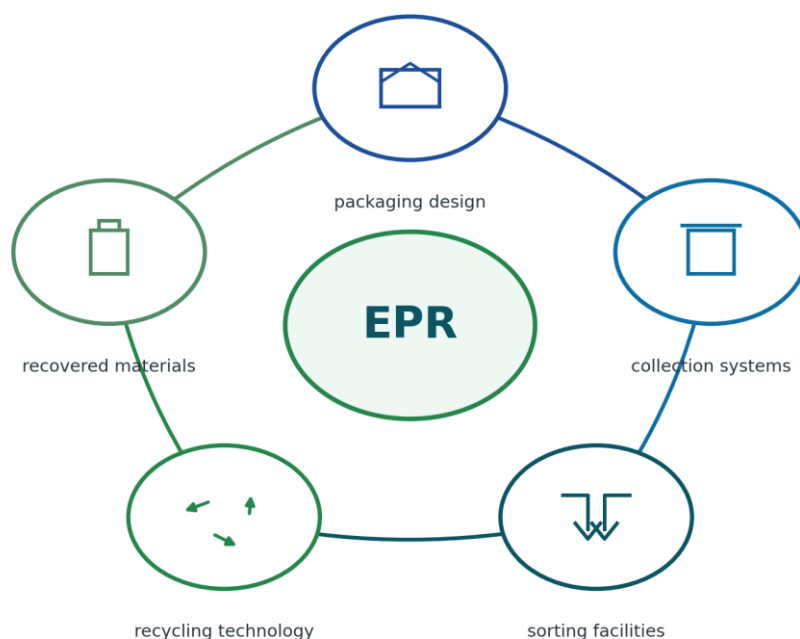
Traditional waste management systems have largely placed responsibility on municipalities and consumers. Under this approach, producers manufacture and sell products, consumers use them, and local authorities manage the waste generated afterward. This creates a disconnect between packaging decisions made during production and the environmental costs created after disposal.

For example, packaging that is inexpensive to manufacture but difficult to recycle may create significant costs for communities and waste systems. EPR seeks to correct this imbalance by ensuring that producers contribute to the systems required to manage packaging after use.

However, effective EPR is not simply a mechanism for collecting producer fees. Its broader purpose is to change behaviour throughout the plastic value chain.

A well-designed EPR system encourages manufacturers to produce packaging that is easier to recycle, repair, or reuse. It supports investment in recovery infrastructure. It creates incentives to reduce unnecessary materials and promotes a circular economy where resources remain in use for longer periods.

The success of EPR should therefore be measured through environmental outcomes, including reductions in plastic leakage, improved collection rates, increased recycling, and greater use of recovered materials.



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Tanzania's Plastic Packaging Governance Context: Progress and Remaining Gaps

Tanzania has increasingly recognized the environmental risks associated with plastic pollution and has introduced several regulatory measures aimed at reducing problematic plastic products and improving environmental management. These efforts demonstrate growing awareness that plastic pollution requires intervention beyond conventional waste disposal approaches.

One of the most visible policy interventions has been the restriction of plastic carrier bags through the Environmental Management (***Prohibition of Plastic Carrier Bags and Plastic Bottle Cap Seals Regulations, 2022***). This measure represented an important step toward reducing unnecessary plastic consumption and improving environmental protection.

However, plastic packaging challenges extend beyond carrier bags. Modern consumer economies rely on a wide range of plastic materials, including beverage bottles, food packaging, household product containers, agricultural packaging, and industrial materials.

Therefore, a comprehensive solution requires systems capable of managing plastic packaging throughout its entire life cycle. Restrictions on specific products can reduce certain waste streams, but they do not replace the need for collection, recycling, recovery, and circular economy systems.

The National Environment Management Council (NEMC) plays a central role in environmental regulation, compliance monitoring, and enforcement, while local government authorities remain responsible for important aspects of municipal solid waste management. This institutional structure provides a foundation for EPR implementation. However, effective plastic packaging management requires stronger coordination between these institutions and private-sector actors.

The main challenge facing Tanzania is therefore not the absence of environmental policy. Rather, the challenge is ensuring that policy commitments translate into measurable reductions in plastic leakage.

An effective EPR system requires clear responsibilities among producers, regulators, local authorities, recyclers, and communities. Without this coordination, responsibility can become fragmented, with each actor addressing only part of the problem while plastic continues to escape into the environment.

Why Existing Frameworks Have Not Yet Delivered Sufficient Reductions in Plastic Leakage

The continued visibility of plastic waste in Tanzania's urban and natural environments suggests that implementation barriers remain significant. These barriers are not unique to Tanzania; many countries with EPR frameworks experience similar difficulties when moving from policy design to practical delivery.

One of the most important barriers is **inadequate collection infrastructure**. Collection is the foundation of any recycling system because materials that are not collected cannot be recovered. Where households, businesses, and institutions lack reliable waste collection services, plastic packaging often enters uncontrolled disposal pathways, including open dumping, burning, drainage channels, and waterways.

This challenge is particularly significant in rapidly growing urban areas. As cities expand and consumption increases, waste management systems must continuously adapt. If collection capacity does not grow at the same pace as waste generation, plastic leakage becomes increasingly difficult to control.

Research on waste management systems in developing countries shows that inadequate collection is often the largest factor contributing to environmental leakage. Improving recycling technology alone cannot solve the problem if plastics never enter formal recovery systems.



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A second major barrier is the **limited economic value of many plastic packaging materials**. Plastic waste is not a single material category. Some plastics have strong market demand because they can be easily recycled into valuable products. Others, such as **multilayer packaging and contaminated flexible plastics**, are technically difficult and economically unattractive to recover.

This creates a market failure. The materials causing significant environmental damage are often those that generate the lowest financial returns for recyclers. Without additional support, private recycling markets will naturally focus on profitable materials and leave low-value plastics unmanaged.

EPR provides a mechanism to correct this imbalance by requiring producers to contribute toward the recovery of all relevant packaging materials, including those that lack sufficient market value.

A third challenge is weak connection between producer responsibility and actual environmental performance. In some EPR systems, companies meet formal obligations by registering, reporting, or paying fees, but the resulting environmental improvements remain limited.

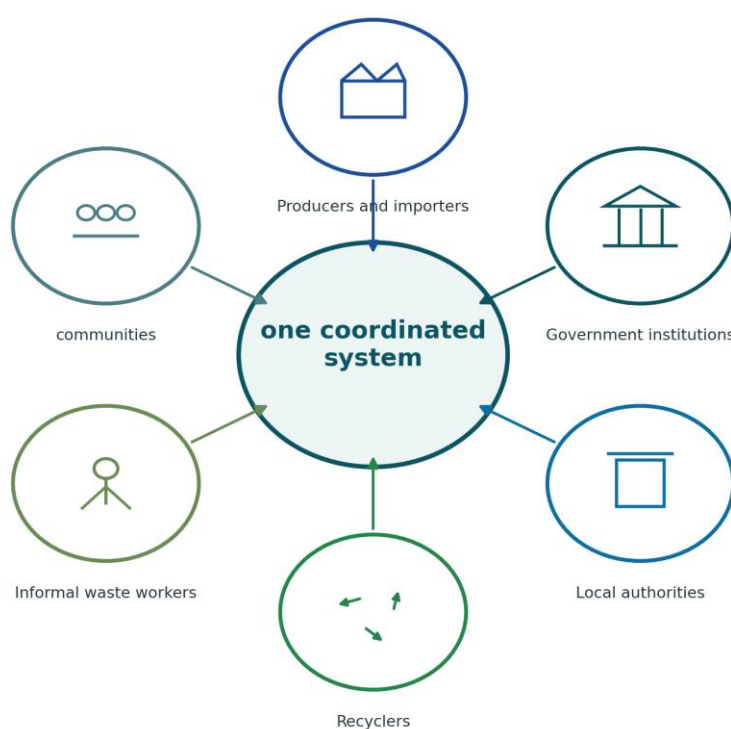
A successful EPR system must therefore move beyond administrative compliance and focus on measurable outcomes. The key question should not only be whether producers are participating, but whether their participation is reducing plastic leakage.

Institutional Reforms Required for Effective EPR Implementation in Tanzania

The effectiveness of plastic packaging EPR depends heavily on institutional coordination. Plastic waste management involves multiple actors with different responsibilities, and weaknesses in coordination can prevent the system from functioning effectively.

Producers and importers introduce packaging into the market and therefore influence the quantity and design of plastic materials entering the economy. Government institutions establish regulations, monitor compliance, and provide oversight. Local authorities manage municipal waste services. Recyclers transform recovered materials into secondary resources. Informal waste workers often collect and recover valuable materials that would otherwise be lost.

For EPR to succeed, these actors must operate as part of one coordinated system rather than as separate activities.



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Tanzania would benefit from a stronger institutional mechanism that links producer obligations with national and local waste management priorities. Producer-funded activities should complement municipal systems rather than operate separately from them.

For example, investments made by producers should consider where plastic leakage occurs, which materials are most problematic, and where collection infrastructure is most needed. Similarly, local governments should have mechanisms for coordinating with Producer Responsibility Organizations and recycling companies.

Clear institutional roles are particularly important because unclear responsibility can result in gaps where no actor assumes ownership of certain activities. Low-value plastic recovery, public awareness, data collection, and monitoring are areas that often require explicit coordination.

Strengthening Monitoring, Reporting, and Enforcement of Producer Obligations

Monitoring and enforcement are essential components of any effective EPR system. Without reliable information, regulators cannot determine whether producers are meeting their obligations or whether the system is achieving environmental improvements.

- ❖ A robust monitoring system should track the amount and types of plastic packaging introduced into the market, the quantities collected, the amounts recycled, and the final destinations of recovered materials.
- ❖ Producer reporting should be supported by independent verification mechanisms. Without auditing, there is a risk that reported recovery achievements may not accurately represent actual environmental performance.
- ❖ Tanzania should also ensure that compliance obligations are meaningful and enforceable. Regulations without effective enforcement mechanisms may create limited behavioural change because producers face insufficient incentives to improve packaging design or invest in recovery systems.
- ❖ At the same time, enforcement should be combined with transparency and technical support. Producers need clear guidelines, reporting systems, and practical pathways for compliance. Regulators require adequate institutional capacity to monitor performance consistently.
- ❖ The goal of enforcement should not simply be punishment. It should ensure that all market participants contribute fairly to reducing plastic leakage.

Financial Reforms: Creating Incentives for Circular Plastic Systems

A sustainable EPR system requires a financing model that supports the full plastic recovery chain. Producer contributions should not be viewed only as compliance costs but as investments in preventing environmental damage and building circular economy infrastructure.

Traditional waste management systems often place significant financial responsibility on municipalities, which may already face limited resources. EPR provides an opportunity to introduce additional financing from those benefiting economically from plastic packaging.

However, financing mechanisms must be carefully designed. If producer fees are too low, they will not support meaningful improvements. If they are poorly structured, they may fail to encourage better packaging choices.

One important reform is the introduction of eco-modulated fees. Under this approach, producers using packaging that is easily recyclable or reusable would face lower contributions, while producers using difficult-to-recycle materials would contribute more. This creates a direct economic incentive for improved packaging design.

For example, a company producing easily recyclable mono-material packaging would receive a different financial signal from a company producing complex multilayer packaging with limited recovery options.

Over time, this approach can influence manufacturing decisions and reduce the amount of problematic plastic entering the market.



Improving Collection, Recycling Infrastructure, and Circular Economy Performance

Infrastructure development is one of the most important requirements for successful EPR implementation. Plastic cannot be recycled if it is not collected, and collected materials cannot create circular value without adequate sorting and processing capacity.

- ✓ *Tanzania requires investment across the entire recovery chain, including collection systems, sorting facilities, transportation networks, recycling technology, and markets for recycled products.*
- ✓ *Infrastructure development should reflect Tanzania's geographic and economic conditions. Large-scale facilities may be appropriate for major urban centres, while decentralized collection and recovery systems may be more practical in smaller towns and rural areas.*
- ✓ *The informal sector should also be considered an important component of this infrastructure. Informal waste collectors already contribute to material recovery, particularly where formal collection systems are limited.*
- ✓ *Rather than replacing these systems, EPR should integrate and strengthen them by improving working conditions, supporting organization, and creating stronger links with recycling companies.*
- ✓ *A circular economy approach requires recognition that waste management is not only an environmental service but also an economic activity capable of creating employment and generating value from recovered materials.*

Policy Recommendations for Tanzania's Plastic Packaging EPR System

The future effectiveness of plastic packaging EPR in Tanzania will depend on shifting from a framework-focused approach toward an outcome-focused system.

- The first priority should be strengthening implementation capacity. Regulations, producer obligations, and institutional arrangements must be supported by effective monitoring and enforcement.
- The second priority should be establishing sustainable financing mechanisms that support collection, recycling, and recovery rather than only administrative compliance.
- The third priority should be improving collection coverage and integrating informal recovery systems into formal EPR structures.
- The fourth priority should be encouraging better packaging design through economic incentives that reward recyclability, reuse, and reduced material use.
- Finally, Tanzania should develop stronger data systems to measure progress. The success of EPR should be assessed through environmental outcomes, including reductions in plastic leakage, increased recycling, improved recovery rates, and cleaner communities.

Conclusion

Plastic packaging will continue to play an important role in Tanzania's economic development. The challenge is therefore not simply eliminating plastic use but creating systems that ensure plastics are responsibly designed, collected, recovered, and reused.

Extended Producer Responsibility provides an opportunity to transform plastic management by connecting producers, governments, recyclers, and communities through shared responsibility.

However, EPR will only succeed if Tanzania moves beyond policy development and focuses on implementation. Effective enforcement, sustainable financing, collection infrastructure, recycling markets, institutional coordination, and reliable monitoring are essential.

Plastic pollution is ultimately a symptom of a system that does not yet fully connect production with environmental responsibility. By strengthening EPR implementation, Tanzania can reduce plastic leakage, protect ecosystems, support public health, and move toward a circular economy where materials remain valuable resources rather than becoming waste.



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Sources

- Gallo, F., Fossi, C., Weber, R., Santillo, D., Sousa, J., Ingram, I., Nadal, A., & Romano, D. (2018). Marine litter plastics and microplastics and their toxic chemicals components: The need for urgent preventive measures. *Environmental Sciences Europe*, 30, 13.
- Leslie, H. A., van Velzen, M. J. M., Brandsma, S. H., Vethaak, A. D., Garcia-Vallejo, J. J., & Lamoree, M. H. (2022). Discovery and quantification of plastic particle pollution in human blood. *Environment International*, 163, 107199.
- Organisation for Economic Co-operation and Development. (2022). *Global Plastics Outlook: Economic Drivers, Environmental Impacts and Policy Options*. OECD Publishing.
- Ragusa, A., Svelato, A., Santacroce, C., Catalano, P., Notarstefano, V., Carnevali, O., Papa, F., Rongioletti, M. C. A., Baiocco, F., Draghi, S., D'Amore, E., Rinaldo, D., Matta, M., & Giorgini, E. (2021). Plasticenta: First evidence of microplastics in human placenta. *Environment International*, 146, 106274.
- United Nations Environment Programme. (2023). *Turning off the Tap: How the World Can End Plastic Pollution and Create a Circular Economy*. UNEP.
- Wilcox, C., Seville, E. V., & Hardesty, B. D. (2015). Threat of plastic pollution to seabirds is global, pervasive, and increasing. *Proceedings of the National Academy of Sciences*, 112(38), 11899–11904.
- World Bank. (2018). *What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050*. World Bank.
- World Bank. (2024). *Tanzania: Reducing methane in waste and livestock sectors*. World Bank.
- National Bureau of Statistics. (2017). *National environment statistics report, 2017: Tanzania Mainland*. Government of the United Republic of Tanzania. NBS Environment Statistics Report 2017
- Daily News. (2025, February 1). *Govt moves to strengthen waste management*. Daily News Tanzania
- International Pollutants Elimination Network. (2026). *Phthalates and bisphenols: Tanzania*.
- Kimaro, O. (2025, June 12). Beyond the bag ban: Time for tougher plastic pollution policies in Tanzania. *The Citizen*.
- The Citizen. (2024, February 20). *Stakeholders rally for plastic bottle recycling to safeguard Tanzania's environment*.
- National Environment Management Council. (2023, May 16). *Katazo la mifuko ya plastiki na hali ya matumizi ya mifuko mbadala*. Government of the United Republic of Tanzania. NEMC plastic bags statistics

